Sahaj Bhadja

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Summary

Software Engineer with 2+ years shipping **Python** services, multi-step **LLM** pipelines, and **AWS**/Spark data pipelines, hardening AI workflows with fallbacks, cost tracking, and accuracy checks, plus research in **phishing** and malware detection.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Hardened async **LLM orchestration** with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Built **data pipelines** across PostgreSQL, SQL Server, and cloud storage with validation, reconciliation checks, and incremental loads, running 5K+ monthly jobs at 99%+ reliability for 500+ users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built **multi-step LLM pipelines** chaining rubric generation, OCR extraction, grading, and feedback across GPT-4o, Claude, and Qwen-VL, serving 10K+ daily requests for 150+ instructors at 90%+ consistency.
- Added timeout-aware model fallback for unreliable OCR, logged per-call tokens and **cost** per feature, and checked grading consistency to keep the non-deterministic workflow accurate while cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **AWS** Glue and S3 ETL that transformed raw JSON, CSV, and Parquet from multiple sources into normalized datasets, boosting pipeline efficiency 68% and cutting storage costs 15%.
- Wrote 20+ **unit and integration test** suites with PyTest and Mockito in nightly Jenkins CI/CD, and optimized REST API integrations with pagination and caching, improving response times 30%.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Built a phishing and **malicious URL detection** pipeline over 650K+ URLs, engineering 21 behavioral indicators such as IP presence, URL length, and suspicious keywords for model classification.
- Evaluated Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and CNN models against each other, reaching 96%+ **detection accuracy** and reporting actionable phishing and malware trend insights.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built hourly **PySpark** jobs orchestrated by **Airflow** on AWS over 50K+ booking and payment events, aggregating peak hours, service demand, daily active users, and revenue for client reporting.
- Built Python Flask multi-currency payment APIs with **fraud checks** and idempotent webhooks for refunds, disputes, and chargebacks, sustaining sub-200ms latency and cutting manual payment handling by 70%.

Skills

Languages: Python, SQL, TypeScript, JavaScript, Java
LLMs & Evaluation: Prompt Chaining, Multi-Model Routing, RAG, Embeddings, LLM Evaluation, OCR, Claude, GPT-4o
Data Pipelines: PySpark, Spark, Databricks, Airflow, AWS Glue, ETL/ELT, Data Validation
Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, SQL Server, MongoDB, Firebase
Cloud & Ops: AWS (S3, RDS, Glue), GCP, Azure, Docker, CI/CD, Jenkins, Sentry, Monitoring
Testing & Security: PyTest, Mockito, Unit and Integration Testing, Fraud Detection, Phishing Detection, JWT, RBAC, Claude Code

Projects

Business Ownership Research Agent | *Python, Vertex AI, Gemini, FastAPI, PostgreSQL, PyTest*

- Designed a Python tool-calling agent on Vertex AI that queries public company registries to reconstruct **beneficial ownership** with citations, scored by an **eval suite** on accuracy, citation validity, cost, and latency.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, SQL Server, Docker*

- Built an AI-assisted SQL workspace on Java and TypeScript supporting 10+ databases, validating non-deterministic schema-aware generated queries through structured execution to reach 82% **accuracy** against relational databases.

Big-Brain Live | *LLaMA-2, FAISS, Embeddings, PostgreSQL, GCP*

- Built a LLaMA-2 **RAG** assistant on GCP with transformer embeddings and FAISS over 50K+ documents, with an incremental ingestion pipeline that cut retrieval latency 40% at 95% query accuracy.